A power company wants to predict the electricity demand in different regions bas past usage, temperature, and time of year. What type of machine learning approach would be suitable? Outline the steps to build the model.

Ans: Requirement: to predict the electricity demand in different region

a. Domain Learning: Machine Learning (Requirement is clear)
b. UnSupervised Learning (It expected to identify the demands in different region)
c. Clustering Model: As the output to cluster the electricity demands based on the variable past usage, temperature and time of year
d. Import data processing libraries
e. Read data set
f. Get the required column in variable X.
g. Plot the data and check the data behaviour
h. Import Standard Scaler library and Standardize the X variable.
i. Import Clustering Algorithm
j. Train the model using fit function
k. Predict the X value to get Labels (clustering groups)
l. Add this predicted labels into existing dataset
m. Save the dataset into CSV file
n. Send the CSV file to stakeholder.

2. An e-commerce platform wants to identify and filter out fake product reviews based on user behavior, review patterns, and sentiment analysis. What type of machine learning approach would be suitable? Outline the steps to build the model.

Ans: to identify and filter out fake product reviews based on user behavior, review patterns, and sentiment analysis

a. Domain Learning: Machine Learning (Requirement is clear,)
b. UnSupervised Learning (to identify and filter out fake product reviews based on user behavior, review patterns, and sentiment analysis)
c. Clustering Model: As the output to identify and filter out fake product reviews based on user behavior, review patterns, and sentiment analysis
d. Import data processing libraries
e. Read data set
f. Get the required column in variable X.
g. Plot the data and check the data behaviour
h. Import Standard Scaler library and Standardize the X variable.
i. Import Clustering Algorithm
j. Train the model using fit function
k. Predict the X value to get Labels (clustering groups)
l. Add this predicted labels into existing dataset
m. Save the dataset into CSV file

n. Send the CSV file to stakeholder.

3. A city traffic department wants to estimate traffic congestion levels based on GPS data, road construction reports, and weather conditions. What type of machine learning approach would be suitable? Outline the steps to build the model.
   Ans: to estimate traffic congestion levels based on GPS data, road construction reports, and weather conditions

   a. Domain Learning: Machine Learning (Requirement is clear,)
   b. UnSupervised Learning (to estimate traffic congestion levels based on GPS data, road construction reports, and weather conditions)
   c. Clustering Model: to estimate traffic congestion levels based on GPS data, road construction reports, and weather conditions.
   d. Import data processing libraries
   e. Read data set
   f. Get the required column in variable X.
   g. Plot the data and check the data behaviour
   h. Import Standard Scaler library and Standardize the X variable.
   i. Import Clustering Algorithm
   j. Train the model using fit function
   k. Predict the X value to get Labels (clustering groups)
   l. Add this predicted labels into existing dataset

m. Save the dataset into CSV fileSend the CSV file to stakeholder.

4. A bank wants to determine whether a loan applicant should be approved or rejected based on income, credit history, and previous loan repayment behavior. What type of machine learning approach would be suitable? Outline the steps to build the model.

   Ans: to determine whether a loan applicant should be approved or rejected based on income, credit history, and previous loan repayment behavior

   a. Domain Learning: Machine Learning (Requirement is clear,)
   b. Supervised Learning (to determine whether a loan applicant should be approved or rejected)
   c. Classification Model: to determine whether a loan applicant should be approved or rejected.
   d. Import data processing libraries
   e. Read data set
   f. Split data into Independent and Dependent.
   g. Use Onehotencoder to convert categorical data into Numerical
   h. Import model_selection library to split Train and Test data
   i. Import Standard Scaler library and Standardize the X variable.
   j. Import Classification Algorithm

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k. Train the model using fit function
l. Predict the Test value
m. Evaluate the model using confusion_matrix and classification_report
n. Check Recall, Precission, F1_score, Accuracy.
o. Based on evaluation matrix and choose the good model
p. Create deployment package
q. Validate the package with test data
r. Deploy the package

5. A factory wants to automatically detect defective products using images from a quality control camera. What type of machine learning approach would be suitable? Outline the steps to build the model.
   Ans; Image processing would be good do in Deep learning.

6. An agricultural company wants to predict the best crops to grow based on soil composition, rainfall, and past harvest data. What type of machine learning approach would be suitable? Outline the steps to build the model.

   Ans: to predict the best crops to grow based on soil composition, rainfall, and past harvest data

   a. Domain Learning: Machine Learning (Requirement is clear,)
   b. UnSupervised Learning (to predict the best crops to grow based on soil composition, rainfall, and past harvest data)
   c. Clustering Model: to predict the best crops to grow based on soil composition, rainfall, and past harvest data.
   d. Import data processing libraries
   e. Read data set
   f. Get the required column in variable X.
   g. Plot the data and check the data behaviour
   h. Import Standard Scaler library and Standardize the X variable.
   i. Import Clustering Algorithm
   j. Train the model using fit function
   k. Predict the X value to get Labels (clustering groups)
   l. Add this predicted labels into existing dataset
   m. Save the dataset into CSV file
   n. Send the CSV file to stakeholder.

7. A business wants to analyze customer responses to different types of advertisements and promotions to determine the most effective marketing strategy. What type of machine learning approach would be suitable? Outline the steps to build the model.
   Ans to analyze customer responses to different types of advertisements and promotions to determine the most effective marketing strategy
   a. Domain Learning: Machine Learning (Requirement is clear,)

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b. UnSupervised Learning (to analyze customer responses to different types of advertisements and promotions to determine the most effective marketing strategy)
c. Clustering Model: to analyze customer responses to different types of advertisements and promotions to determine the most effective marketing strategy
d. Import data processing libraries
e. Read data set
f. Get the required column in variable X.
g. Plot the data and check the data behaviour
h. Import Standard Scaler library and Standardize the X variable.
i. Import Clustering Algorithm
j. Train the model using fit function
k. Predict the X value to get Labels (clustering groups)
l. Add this predicted labels into existing dataset
m. Save the dataset into CSV file
n. Send the CSV file to stakeholder.

8. A software company wants to build a system that can analyze source code and predict whether a particular piece of code is likely to contain a bug. What type of machine learning approach would be suitable? Outline the steps to build the model.
Ans to build a system that can analyze source code and predict whether a particular piece of code is likely to contain a bug
    a. Domain Learning: Machine Learning (Requirement is clear,)
    b. UnSupervised Learning (to build a system that can analyze source code and predict whether a particular piece of code is likely to contain a bug)
    c. Clustering Model: to build a system that can analyze source code and predict whether a particular piece of code is likely to contain a bug
    d. Import data processing libraries
    e. Read data set
    f. Get the required column in variable X.
    g. Plot the data and check the data behaviour
    h. Import Standard Scaler library and Standardize the X variable.
    i. Import Clustering Algorithm
    j. Train the model using fit function
    k. Predict the X value to get Labels (clustering groups)
    l. Add this predicted labels into existing dataset
    m. Save the dataset into CSV file
    n. Send the CSV file to stakeholder.

9. A fitness app wants to recommend personalized workout plans for users based on their exercise history, fitness level, and preferences. What type of machine learning approach would be suitable? Outline the steps to build the model.

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Ans to recommend personalized workout plans for users based on their exercise history, fitness level, and preferences

a. Domain Learning: Machine Learning (Requirement is clear,)
b. UnSupervised Learning (to recommend personalized workout plans for users based on their exercise history, fitness level, and preferences)
c. Clustering Model: to recommend personalized workout plans for users based on their exercise history, fitness level, and preferences
d. Import data processing libraries
e. Read data set
f. Get the required column in variable X.
g. Plot the data and check the data behaviour
h. Import Standard Scaler library and Standardize the X variable.
i. Import Clustering Algorithm
j. Train the model using fit function
k. Predict the X value to get Labels (clustering groups)
l. Add this predicted labels into existing dataset
m. Save the dataset into CSV file
n. Send the CSV file to stakeholder.

9 A social media platform wants to detect fake accounts by analyzing user activity, posting patterns, and interactions with other users.Give steps to Achieve.

Ans to detect fake accounts by analyzing user activity, posting patterns, and interactions with other users.Give steps to Achieve.

a. Domain Learning: Machine Learning (Requirement is clear,)
b. Supervised Learning (to detect fake accounts)
c. Classification Model: to detect fake accounts.
d. Import data processing libraries
e. Read data set
f. Split data into Independent and Dependent.
g. Use Onehotencoder to convert categorical data into Numerical
h. Import model_selection library to split Train and Test data
i. Import Standard Scaler library and Standardize the X variable.
j. Import Classification Algorithm
k. Train the model using fit function
l. Predict the Test value
m. Evaluate the model using confusion_matrix and classification_report
n. Check Recall, Precission, F1_score, Accuracy.
o. Based on evaluation matrix and choose the good model
p. Create deployment package
q. Validate the package with test data
r. Deploy the package

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